Whose Fairness? Auditing Exposure Fairness When AI Agents Read Rankings

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Abstract

Exposure fairness is not observed directly; it is computed using an examination curve that weights rank positions and is conventionally treated as a single known measurement instrument. AI agent readers challenge this assumption. In a bias-controlled study of eight LLMs across six families, agent readers exhibit distinct positional curves, including mild primacy, near-flat behavior, mid-list humps, and reversed primacy. No agent curve matches the human model. This setting involves several coexisting curves rather than uncertainty about a single curve, making the instrument set-valued and exposure fairness only partially identified. Under standard existence-style auditing, a ranking can be certified by a plausible curve even when every realized reader is treated unfairly. On one workload, calibration to the human curve produces greater unfairness for the worst-served agent reader than no intervention. We characterize this failure, show that the exact worst case and certificate expiry are computable from the set, and formulate a min–max robust ranking policy as a single linear program. Across four domains, an oracle given the realized curve incurs about a quarter of the fairness premium; the remainder is attributable to serving all readers simultaneously. Robustness costs at most 0.005 NDCG beyond human-calibrated fairness. Residual ambiguity remains at the individual-provider level. For 27–73% of the top-15, the direction of exposure harm is unidentified, although this disagreement is averaged out in a group-level audit. On LastFM, changing the aggregation level produces a thirty-fold gap.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; *Evaluation of retrieval results*; • **Computing methodologies** → Artificial intelligence.

Keywords

exposure fairness; position bias; examination model; partial identification; robust ranking; AI agent; fairness auditing

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1 Introduction

Ranking systems allocate attention, and exposure-fairness audits ask whether that allocation is proportional to provider merit [6, 10, 30]. The audited quantity is not observed directly. An *examination curve* v weights each position to convert rank into exposure. The curve therefore serves as the measurement instrument defining exposure. Classical systems estimate one such curve from interaction logs [2, 16] and downstream analyses proceed as if it were fixed, despite recognized uncertainty in position-bias estimation [14, 24].

AI agent readers challenge the assumption of a single examination curve. Rankings are increasingly consumed by agentic assistants and LLM-mediated search [4], and the automated share of Web requests is already large but imperfectly observed [1, 9, 36], so an auditor can assume only an interval. These readers also differ in their positional responses: prior work reports model- and task-dependent position effects [20, 29], and our controlled measurements recover mild primacy, near-flat behavior, a mid-list hump, and reversed primacy across model families. The human curve therefore cannot be replaced by a single agent curve.

This heterogeneity affects the validity of the audit logic, beyond the difficulty of estimating a curve. With one population and one latent curve the uncertainty is epistemic, and asking whether some plausible curve makes a ranking fair is coherent [28]. This criterion is unsound when several reader types coexist, because the multiplicity is then ontic: Theorem 3.1 shows that an existence audit can certify a curve used by no realized reader while every realized type exceeds the audit threshold. Exposure fairness is then partially identified, so a valid audit requires a worst-case certificate over the realized set rather than a point estimate.

A two-item instance illustrates this failure. Let the merits be 0.6 and 0.4, and let humans read with positional weights $(0.7, 0.3)$ while one agent type uses $(0.3, 0.7)$, an increasing curve of the kind we measure in three model families. The identity ranking is exactly merit-proportional under the intermediate curve $(0.6, 0.4)$, so an existence audit over decaying curves certifies it, yet no reader uses that curve: humans experience a worst-provider deviation of 0.25 and the agent 0.75. That curve is precisely the aggregate mixture at a 25% agent share, so calibrating to true average traffic does not repair the audit, and widening the curve family makes certification easier.

We formalize exposure auditing when human and heterogeneous agent readers coexist, examining verdict decidability, the cost of protecting all readers, and the providers affected by residual ambiguity. An oracle given the realized curve incurs only a small part of the premium: 74–100% of it comes from serving heterogeneous readers at once. Residual ambiguity is concentrated at the individual-provider level, where the direction of exposure harm can remain unidentified even when a coarse group audit certifies the platform.

The contributions answer these three questions in turn.

- **Which verdicts survive.** An existence-style certificate can be based on a curve used by no realized reader while every realized reader is treated unfairly. Widening the auditor's family makes

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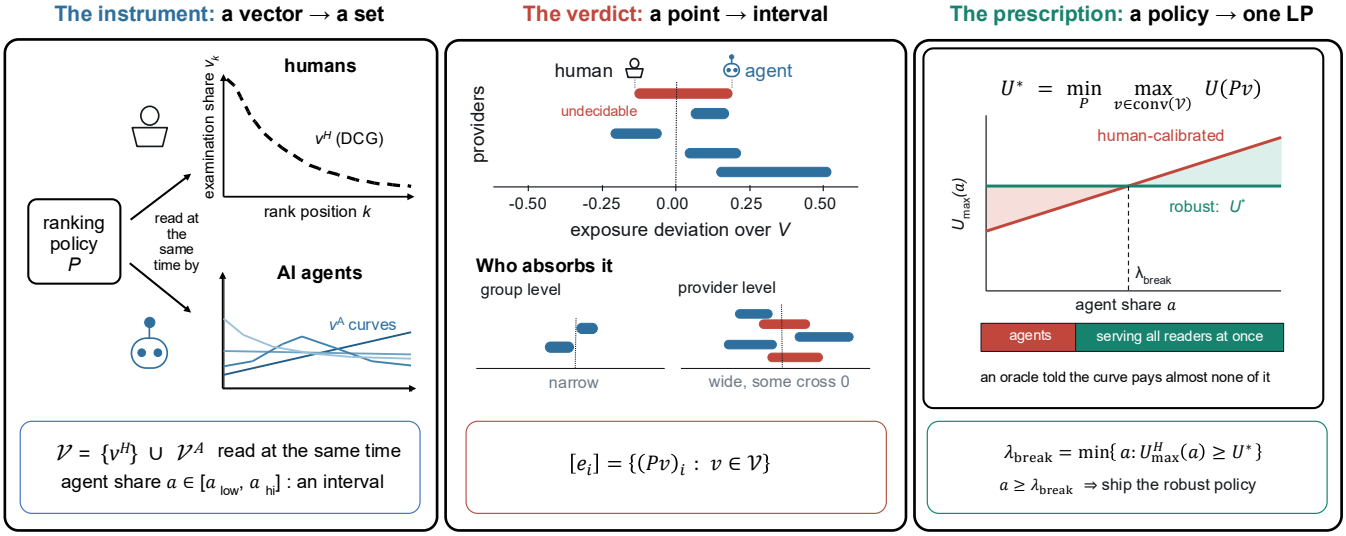


Figure 1: Overview. (Left) Humans and heterogeneous AI agents read the same ranking under different measured curves, so the exposure-fairness instrument is a set $\mathcal{V}$. (Center) Each provider's verdict becomes an interval under $\mathcal{V}$, and some intervals cross zero. This ambiguity affects individual providers rather than coarse groups. (Right) An existence audit can certify a curve used by no realized reader. A single min–max LP instead yields a robust policy that dominates the human-calibrated optimum beyond λ_{break}.

the test trivial without resolving this failure. The worst case over the set is exactly computable, with explicit certificate expiry in agent share and measurement age.

- **What protection costs.** A bias-controlled study of eight LLMs across six families provides an empirical stress-test set. Under this set, 74–100% of the fairness premium is attributable to serving all curves simultaneously rather than to agent presence, and a policy calibrated to the human curve leaves the worst-served agent reader worse off than relevance sorting. One min–max LP protects the whole set for at most 0.005 NDCG.
- **Who absorbs the residual.** Residual ambiguity persists under every policy and affects individual providers, whose direction of exposure harm is unidentified for 27–73% of the top-15. This disagreement is averaged out in a coarse group audit.

2 Related Work

Exposure fairness. Exposure fairness is commonly formulated through stochastic rankings and position-weighted exposure [6, 10, 30]. Two-sided, expected-exposure, and fair-division variants retain the same dependence on a browsing or examination model [11, 27, 28], as do the surveys [24, 38]. Position bias is estimated from clicks through intervention or harvesting [2, 16], and uncertainty in that estimate is a recognized difficulty [14, 24]. The downstream audit nevertheless uses one positional weighting. Even richer browsing-model families instantiate a particular model when fairness is computed [19].

The closest audit logic is Viable-Λ [28], which asks whether some monotone-decay curve makes a ranking fair. This is appropriate for epistemic uncertainty about one population with one latent curve. Our regime is also ontic because several reader curves are realized at the same time. A curve that supports certification may therefore correspond to no reader, and widening the family makes certification easier without resolving this mismatch. The distinction concerns the coexistence of several instruments rather than the size of a confidence region around one unknown instrument.

Robust ranking. Prior fairness work places uncertainty on merits [25, 31], protected attributes [17], or exposure dependencies [14]. Distributionally robust ranking perturbs data distributions, propensities, or user subgroups [23]. We adopt the partial-identification language of sharp sets and set-contingent conclusions [21], but place uncertainty on the positional weighting that defines exposure. Our adversary therefore chooses the measurement instrument rather than data supplied to a fixed metric. Traffic-aware provider-fairness methods [37] address the amount of arriving traffic while retaining one exposure model. Knowing the total agent share does not resolve this ambiguity because one share can contain several agent types whose curves imply different provider exposures. Statistical auditing [8] quantifies sampling uncertainty around a specified fairness quantity, whereas we ask whether that quantity is point-identified at all. Work on how exposure should be valued once known [27] also takes the weighting as given. We instead examine how different readers induce different valid metrics on the same ranking.

Agent consumers. LLM position sensitivity is documented in judging, long-context use, and listwise ranking [20, 29, 32], and LLM rankers have their own fairness profiles [33]. The effects on exposure have also been measured directly: ACES [4] connects shopping-agent position bias to market outcomes, LLM product and news systems redistribute attention across brands, outlets, topics, and sources [5, 18], and generative-search work shows machine-mediated exposure departing from rank-only assumptions and inducing producer-side optimization [3]. Agentic-IR work questions whether human user models transfer to agent readers [39]. These

studies establish heterogeneous behavior under particular readers. We incorporate that heterogeneity into exposure measurement, asking which verdicts remain sound and which policy satisfies them when several instruments coexist.

3 Theory

3.1 Setup

Items $i = 1, \dots, n$ carry merits $m \in \mathbb{R}^n_{>0}$. A stochastic ranking policy induces a doubly stochastic matrix P, where P_{ik} is the probability that item i occupies rank k. By Birkhoff–von Neumann, every such P is implementable. A curve $v \in \mathbb{R}^n_{\geq 0}$, normalized so that $\sum_k v_k = 1$, gives exposure $e = Pv$. A policy is fair when exposure is merit-proportional:

$$e_i \;=\; c\, m_i, \qquad c \;=\; 1 \Big/ \sum\nolimits_i m_i. \tag{1}$$

The normalization makes the fairness target independent of the curve's shape. The shape determines whether that target is reachable. For a convex unfairness measure $U(e)$, the policy tables report the worst-provider norm

$$U_\infty(e) \;=\; \max_i \Big| e_i/(c\, m_i) - 1 \Big|, \tag{2}$$

with a fixed-normalization mean-absolute-deviation (MAD) index as tie-break and width measure.[1]

Agent curves. We use examination curve generically for the positional weight of the reader's value-conferring event, either human examination or agent selection. When an agent selects directly from a list, no provider-observable event separates examination from selection, so v is its position-conditioned choice curve. The measured curves are choice distributions, and the human curve is the DCG examination model [15], $v^H_k = \big(1/\log_2(k{+}1)\big) \big/ \sum_{j=1}^{K} 1/\log_2(j{+}1)$ for $k = 1, \dots, K$, renormalized over the same K positions to match. Reader types are therefore assigned equal value here. The choice κ=1 is normative rather than a normalization, and Section 7 examines alternative valuations.

Ambiguity set. Consumers are heterogeneous. A human-side set $\mathcal{V}^H$ of position-bias models and an agent-side set $\mathcal{V}^A = \{v^{A_1}, \dots, v^{A_J}\}$ are realized at the same time in proportions known only as an interval. Writing a for the agent share, the ambiguity set and its induced mixtures are

$$\mathcal{V} = \mathcal{V}^H \cup \mathcal{V}^A, \quad \mathcal{A}(a) = \big\{(1-a)v^H + av : v \in \mathrm{conv}(\mathcal{V}^A)\big\}. \tag{3}$$

The human and agent sides are treated identically, although most of the observed heterogeneity is on the agent side. The theory is stated at item level. The experiments use group-level targets amortized over queries, $e_g = \sum_q \sum_{i \in g} (P_q v)_i$. Every result below is linear in P, so each applies verbatim under either reading.

Reported quantities. U^*_H and U^* are the minimum worst-provider unfairness attainable under the human curve alone and under all of $\mathcal{V}$. Their difference is the premium, or price of ambiguity. U^*_{orc} is the worst-type value of a policy optimized separately for each realized curve. Thus $U^*_{\mathrm{orc}} - U^*_H$ measures the cost of agent presence, while premium $-$ $(U^*_{\mathrm{orc}} - U^*_H)$ isolates the cost of serving curves simultaneously. The two reported agent shares have different meanings. λ_{break} marks when the human-calibrated policy becomes worse than the robust floor, while $\lambda^*_{\mathrm{cert}}(\tau)$ marks when a certificate at tolerance τ expires. A provider is sign-indeterminate when its deviation interval over $\mathcal{V}$ crosses zero.

[1] Both are convex in e. The classical Gini, normalized by the empirical mean, is not, and is used descriptively only.

3.2 Verdict Logic

When several curves coexist, the audit criterion changes. For a single unknown curve, an existence audit asks whether any plausible curve makes the ranking fair. With several readers present simultaneously, this criterion does not establish fairness across readers.

Theorem 3.1 (False acquittal under simultaneous readers). *Let $\mathcal{V}$ be the realized curve set and D the auditor's parametric family. There exist instances $(m, P, \mathcal{V}, D)$ in which some $v^* \in D \setminus \mathcal{V}$ renders P exactly fair, so an existence audit acquits, while every realized consumer type experiences unfairness above the audit threshold. Widening D does not repair the audit but trivializes it. Acquittal is monotone in D, and at $D = \Delta^{n-1}$ the audit passes whenever some curve rationalizes the policy, irrespective of any realized consumer.*

Proof (constructive, $n = 2$). Take $m = (0.6, 0.4)$, $\mathcal{V} = \{v^H = (0.7, 0.3),\ v^A = (0.3, 0.7)\}$ with v^A increasing, hence outside any monotone decay class D, and $v^* = (0.6, 0.4) \in D$. The unique doubly stochastic P with $Pv^* \propto m$ is the identity, so the existence audit sees $U = 0$ and acquits. Yet $U_\infty(Iv^H) = 0.25$ and $U_\infty(Iv^A) = 0.75$. For any tolerance $\tau < 0.25$, the audit passes a policy under which both realized types are treated unfairly. Here $v^* = 0.75\, v^H + 0.25\, v^A$ is exactly the realized blend at a 25% agent share, so even an audit calibrated to the true aggregate mixture acquits. □

The measurements in Section 4 include three reversed-primacy curves from three model families. The mismatch is typical whenever D axiomatizes monotone decay while $\mathcal{V}$ contains non-monotone curves.

Proposition 3.2 (One computation, two guarantees). *Let the realized types have curves $\{v_t\} = \mathcal{V}$. For convex U and any mixture weights w over them,*

$$U\big(P \sum\nolimits_t w_t v_t\big) \;\leq\; \max_{v \in \mathrm{conv}(\mathcal{V})} U(Pv) \;=\; \max_t U(Pv_t),$$

with strict inequality in general. Hence $\max_t U(Pv_t) \leq \tau$ gives two guarantees simultaneously: (i) $U \leq \tau$ for any realized curve or mixture, and (ii) no consumer type experiences unfairness above τ. The first is epistemic and the second is per-reader, or ontic. Aggregate fairness at the realized mixture does not imply (ii).

An existence audit cannot provide the second guarantee at any tolerance.

Feasibility. A fair policy for a single curve exists iff $c\, m \prec v$ by majorization. Flatter curves shrink the feasible merit set, and in the flat limit changing the ranking no longer changes exposure. Joint feasibility under several curves is one LP feasibility test.

3.3 Computing the Worst Case

Theorem 3.3 (Vertex reduction). *For convex U and a fixed policy P, the partial-identification upper bound over the mixture set $\mathcal{A}(a)$,*

$$U_{\max}(a) \;=\; \max_{v \in \mathcal{A}(a)} U(Pv),$$

is attained at a vertex and is therefore exactly computable by enumerating the measured curves. It is also convex in a. For polyhedral convex U, including U_∞ and MAD, $U_{\max}$ is piecewise linear in a. Whenever $U_{\max}(0) \le \tau$, the critical share $\lambda^*_{\text{cert}}(\tau) = \max\{a : U_{\max}(a) \le \tau\}$ is exactly computable. Convexity ensures that the certificate holds throughout $[0, \lambda^*_{\text{cert}}]$.

The worst case is therefore computable over the specified ambiguity set. Measured curves can change over time, so a certificate must also state how much drift it tolerates.

Proposition 3.4 (Certificates expire on two axes). *Let $\widetilde{\mathcal{V}} = \mathcal{V}^H \cup \widetilde{\mathcal{V}}^A$ re-measure the agent side alone, pairing each $v \in \mathcal{V}^A$ with a $\tilde{v} \in \widetilde{\mathcal{V}}^A$ at total variation at most ε, and let $L = \max_i 1/(c\, m_i)$. Then, for every policy P and every such pair, $\left|U_\infty(Pv) - U_\infty(P\tilde{v})\right| \le L\varepsilon$, and, with U^* the robust value,*

$$\left|U^*(\mathcal{V}) - U^*(\widetilde{\mathcal{V}})\right| \le L\varepsilon, \qquad \left|U_{\max}(a;\mathcal{V}) - U_{\max}(a;\widetilde{\mathcal{V}})\right| \le La\varepsilon,$$

the share a appearing in the last bound because the human curve is held fixed, so the perturbation reaches $\mathcal{A}(a)$ only through its agent component. A certificate $\lambda^*_{\text{cert}}(\tau)$ issued on $\mathcal{V}$ therefore remains valid on $\widetilde{\mathcal{V}}$ up to share $\lambda^*_{\text{cert}}\big(\tau - L\lambda^*_{\text{cert}}(\tau)\varepsilon\big)$. This is conservative because it charges the perturbation at the old expiry share. To first order the expiry shrinks by $L\lambda^*_{\text{cert}}\varepsilon/s$, where $s > 0$ is the slope of $U_{\max}$ at λ^*_{cert}.

Proof. The difference $\delta = v - \tilde{v}$ sums to zero, with positive and negative parts each of mass at most ε. Since $P_{ik} \in [0,1]$, we have $|(P\delta)_i| = |\sum_k P_{ik}\delta_k^+ - \sum_k P_{ik}\delta_k^-| \le \varepsilon$. Dividing by $c\, m_i$ and maximizing over i gives the first bound. The minimum over P and the maximum over $\mathcal{A}(a)$ both preserve it, and the latter contracts the perturbation by a because $(1-a)v^H$ is common to both sets. Piecewise linearity from Theorem 3.3 gives the first-order form. □

The observed changes in the empirical floor are much smaller than this worst-case Lipschitz bound. Both the agent share and the age of the set remain observable audit inputs.

3.4 Prescription

Theorem 3.5 (Robust fair policy). *For polyhedral convex U (as U_∞ and MAD throughout),*

$$U^* = \min_P \max_{v \in \text{conv}(\mathcal{V})} U(Pv)$$

is attained by the solution of a single LP with one constraint block per measured curve, since the inner maximum is attained at vertices by Theorem 3.3. Moreover, U^* is a lower bound on the worst-case unfairness attainable by any policy.

If reading is not separable in content and position, replace Pv by the reader-specific kernel $e_i = \sum_k P_{ik} W_{ik}$; feasibility, vertex reduction, and the robust LP are unchanged because each is linear in P.

4 Measuring Reader Heterogeneity

The theory requires that readers coexisting on the same ranking induce distinct positional weights. Position bias in LLMs, including open models, is well documented [4, 20, 29, 32, 40]. An exposure-fairness audit needs a per-rank curve over a full results page, measured under controlled randomization and normalized so that readers enter the same fairness calculation. We measure these curves and test whether heterogeneity persists under changes in readout, interface, list length, and item domain.

Scope. The theory depends on the coexistence of distinct reader curves, not on any particular curve. The set's members determine certificate expiry and which provider verdicts become sign-indeterminate, whereas the underlying allocative conflict largely determines the premium. Human heterogeneity does not substitute for the agent side. Standard human position-bias models are all monotone decay, and including them as a set has little effect on the robust floor (Table 4). Agent curves add non-monotonicity, the condition under which a monotone-decay existence audit fails.

Protocol. Each trial is a full results page of K=20 synthetic product items, with item–position assignment randomized so that position is independent of quality and the letter–position mapping redrawn per trial to average out letter priors [40]. We read the choice distribution from next-token logits over the label tokens, which gives complete compliance without sampling noise; for API models we use first-token log-probabilities where available and otherwise a sampling protocol validated against logit readout. Two tasks are used: pick defines the preference curve v^A, and cheapest has ground truth. We measure eight models across six families (Table 1). Policies use the deployed K=10 by truncating and renormalizing the K=20 curves, with direct K=10 re-measurement as a robustness check. Serialization is part of the instrument: with explicit rank markers the curves are anchored to position, whereas without them Phi-4-mini follows context order (its forward and reversed presentations agree at $r = 0.90$ once one is reversed), so every curve is measured under one disclosed serialization. Serialization, model selection, and prompts accompany the released bundle.

Findings. The main finding is heterogeneity across models rather than a particular curve shape. Four shape classes appear across the eight models (Figure 2a, Table 1). They are mild primacy, near-flat behavior, a mid-list hump, and reversed primacy in three models from three families. Task incompetence does not account for these shapes: the GPT tiers and Qwen3-VL-32B solve the instrumental task almost perfectly while exhibiting some of the least top-heavy curves, and reversed primacy survives quantization, although for Llama-3.1 competence and position bias cannot be fully separated. In (top-3 mass, r_H) space no agent curve is close to the human model (Figure 2b).

Stress tests. Our curves are position-conditioned choice distributions, and randomizing content against position isolates the positional factor just as randomization-based estimation does for human clicks [2, 16]. All tested conditions change the set's membership, but heterogeneity remains in every case (Table 4). Logit and sampling readouts agree on identical trials, and a two-step tool-using agent on real ESCI queries makes choices similar to those in its single-shot setting. Counterfactual probes show broad access and more concentrated choices, and rebuilding the set from access curves increases the premium, so the result does not depend on identifying choice with examination. Direct K=10 re-measurement moves every curve toward the human model yet raises the premium in all four domains, so truncation was conservative, although the robust policy must be solved on the set actually measured. Re-measuring seven of the eight models on real ESCI queries and product titles likewise reduces the differences among curves without eliminating them: every curve moves toward the human model,

Table 1: The main-set ambiguity set: eight LLMs measured at K=20 (synthetic items, pick task; v^A = choice distribution over ranks). Top-3 = choice mass on ranks 1–3 (human DCG model: 0.30); argmax = rank with the largest share; r_H = correlation with the human curve; cheapest = instrumental-task accuracy. Direct K=10 re-measurement: Table 4.

Model	Family	Size, precision	Protocol	Top-3	argmax rank	r_H	Cheapest	Shape
Qwen3-VL-8B	Qwen	8B, bf16	logit	0.14	3	-0.24	0.72	near-flat, low top
Qwen3-VL-32B	Qwen	32B, 4-bit	logit	0.03	20	-0.73	0.90	reversed primacy
Gemma-4-12B	Gemma	12B, nf4	logit	0.14	5	-0.00	0.94	mid-list hump
Llama-3.1-8B	Llama	8B, bf16	logit	0.04	7	-0.69	0.50	reversed primacy
Mistral-Small-3.2	Mistral	24B, nf4	logit	0.15	7	+0.30	0.71	near-flat
Phi-4-mini	Phi	3.8B, bf16	logit	0.12	7	-0.13	0.42	mid-list hump
GPT-4o	GPT	API	logprob	0.11	9	-0.34	0.97	near-flat
GPT-5.6-sol	GPT	API	sampling	0.14	7	-0.19	1.00	near-flat

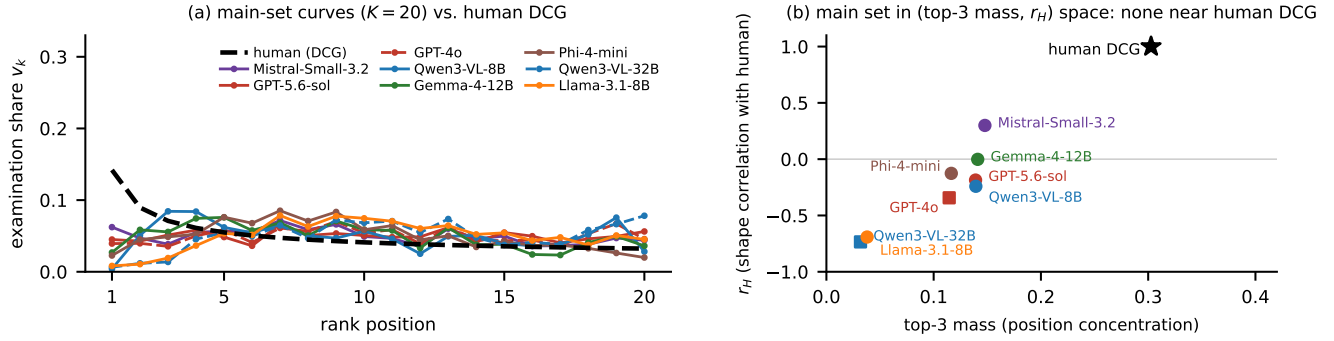


Figure 2: The measured curves. (a) The eight main-set curves (K=20) and the human DCG model. No agent curve matches the human model or another agent curve. (b) The same eight curves in (top-3 mass, r_H) space; the star marks human DCG. Every agent curve is far less top-heavy than the human model, and their shape correlations with it range from −0.73 to +0.30.

reversed and top-heavy shapes still coexist, and re-solving every LP keeps each domain's premium within a few hundredths of its value while increasing the break-even share. Distinct curves therefore persist under realistic items, as required for the audit argument.

5 Experiments

5.1 Cross-Domain Setup

Datasets. The four domains use corporate and demographic definitions of a provider. Amazon ESCI [26] uses US queries, brands as providers, and relevance labels as merit. TREC Fair Ranking 2022 [12] uses sub-continental regions and predicted article quality. Steam uses the P-MMF preprocessing [34, 35], with 43 game publishers and matrix-factorization scores. LastFM-360K [7] uses artists with a real demographic attribute [13] and log play counts. ESCI is a constructed testbed whose brand grouping is ours, and it is also the geometrically hardest domain, so it is used for the detailed analyses while every main claim is replicated on all four domains, with one canonical 200-query workload per domain and 20-workload bootstrap suites where stated.

Metrics. We report the worst-provider norm U_∞ because it provides a worst-case guarantee over every consumer type and provider. TREC's AWRF [12, 28] is an averaged divergence that inherits the same set-valuedness; Table 3 (lower block) repeats the policy comparison under MAD, AWRF, and Gini. All policies use $\mathcal{V}^H = \{v^H\}$, the eight-agent set, and the lexicographic optimum, which minimizes U_∞ and then MAD under v^H. Each 200-query LP has 20,000 rank-marginal variables, 4,000 doubly stochastic equalities, and 30 deviation rows per curve block; HiGHS solves the nine-block robust LP in under ten seconds. We will release code, curves, and workload manifests. An anonymized bundle accompanies the submission.[2]

Cross-domain regimes. We associate examination curves with readers rather than corpora, so we hold the measured set fixed across domains. The decomposition $U^* = U^*_H$ + premium separates geometric hardness from the premium, and the domains fall into three regimes. On ESCI, fairness is geometrically hard even under the known curve and ambiguity adds little. On TREC, Steam, and LastFM, exact fairness is achievable under the known curve, so the entire robust floor is the premium. It is large on TREC and LastFM but near zero on Steam. Steam has the smallest premium, yet its certificates expire earliest, sign-indeterminacy is among the highest, and the robust policy does not reduce it. We conjecture the cause is Steam's geometry: 43 publishers with nearly tied matrix-factorization scores compete for ten slots per user, so exact fairness has low cost under any single curve, whereas small curve differences can reverse many providers' deviation signs. The premium

[2]https://anonymous.4open.science/r/Whose-Fairness-Auditing-Exposure-Fairness-When-AI-Agents-Read-Rankings-0350

Table 2: Main results: partial identification and its price, four domains (top-15 providers, K=10; canonical workload, brackets = 5–95% band over 20 bootstrap workloads; the ESCI λ bracket is the range over workloads with positive premium). U_H^* / U^* = human-only / robust floor; premium = $U^* - U_H^*$; $U_{\text{orc}}^* - U_H^*$ = the premium an oracle told the realized curve would still pay (agent *presence*); the rest (74–100% everywhere) is the price of simultaneity; λ_{break} = break-even agent share of the canonical human-calibrated optimum, and under the robust tie-break (RTB, §5.4); s-ind. = sign-indeterminate providers of 15, human-calibrated / robust; worst type = human-calibrated policy under its worst agent curve; NDCG$_H$ / Δ_{rob} = NDCG@10 of the human-calibrated fair policy / extra cost of the robust policy.

Domain (provider)	U_H^*	U^*	premium [5–95%]	$U_{\text{orc}}^* - U_H^*$	λ_{break} [5–95%]	RTB	s-ind. H / R	worst type	NDCG$_H$ / Δ_{rob}
ESCI (brand)	.192	.231	.039 [.00,.09]	.007	.42 [.32,.48]	.48	4 / 2	.530	.936 / .005
TREC'22 (region)	.000	.181	.181 [.15,.27]	.021	.33 [.31,.38]	.56	11 / 10	.555	.887 / .001
Steam (publisher)	.000	.031	.031 [.02,.10]	.000	.13 [.12,.34]	.59	11 / 14	.250	.931 / .005
LastFM (artist)	.000	.209	.209 [.13,.23]	.053	.42 [.28,.44]	.58	10 / 5	.498	.968 / .001

measures the cost of optimization, whereas indeterminacy describes the limits of the audit. The two quantities need not change together, so we examine cost and incidence separately below.

5.2 Decidable Verdicts

Existence audits. Verdicts are issued over workloads, not single queries. On a ten-item page, the worst-case audit certifies no rankings at any $\tau < 1$ because the page cannot be merit-proportional for every curve at once. Across the 20 bootstrap workloads, existence audits certify a strict superset of the workloads certified by the worst-case audit at every tolerance (Figure 3b). All of those certificates fail the worst-case audit up to $\tau = 0.28$. The failure rate falls to 95% and then 70% at the loosest tolerances. The comparison is nontrivial: the worst-case audit certifies one workload in twenty at $\tau \in [0.30, 0.34]$ and six beyond. This pattern is not specific to the geometric family (Figure 3b). Narrow families issue false certificates, whereas wide families certify every workload. The worst-case guarantee avoids both failures. The same pattern appears query by query without amortization: a Viable-Λ-style search over geometric-decay curves certifies 27/200 queries at $\tau \le 0.25$, all of which fail the worst-case audit. This reproduces the failure described in Theorem 3.1 under the deployed human-calibrated policy.

Certification frontier. The known-curve and robust floors must be distinguished. On ESCI, certification at $\tau = 0.1$ is impossible even under the known curve because $U_H^* = 0.192$. Over the ambiguity set, the floor rises to $U^* = 0.231$, and the difference is the premium. Across 20 bootstrap workloads, both floors are stable but their difference is not. The premium ranges from 0 to 0.137 and vanishes on six geometrically hard workloads. Wherever it is positive, λ_{break} lies between 0.32 and 0.48.

The frontier in Figure 3a shows the range of tolerances for which ambiguity changes the verdict. Outside $\tau \in [0.18, 0.32]$, all workloads receive the same verdict with or without curve knowledge. Within that interval, verdicts differ. At $\tau = 0.20$, 40% of workloads certify when the curve is known and none certify under ambiguity. For a platform using the human-calibrated policy, $\lambda_{\text{cert}}^*(\tau)$ expresses tolerance as an agent-share limit (Figure 3c). Within this interval, each 0.01 increase in tolerance allows roughly three to four additional percentage points of agent share. The band across workloads is wide, and the choice of deployed LP optimum shifts the curve as much as the workload does (§5.4).

Table 3: Upper bound $U_{\max}(a)$ at agent share a and worst-type unfairness (attaining curve), ESCI. The share-agnostic robust policy (Thm. 3.5) guards every $a \in [0, 1]$; the share-bounded variant solves the same LP over $\text{conv}(\{v^H\} \cup \mathcal{A}(0.25))$ and therefore guards $a \le 0.25$ only. †A distinct policy whose reported values coincide with the human-calibrated optimum (§5.3). Lower block: the human-calibrated and robust policies re-solved and evaluated under other indices (MAD on ESCI; AWRF on TREC'22; Gini of the top-15 on Steam), ‡P-MMF [34] audited post hoc on the canonical Steam workload, not ESCI, on its own metric, the minimum provider quota share (0.60 on position-free count exposure).

Policy	a=0	a=0.25	worst type
Relevance sort	.287	.286	.464 (Llama)
Human-calibrated	**.192**	**.213**	.530 (Qwen3-VL-32B)
Average-curve	.327	.304	.353 (Qwen3-VL-8B)
Mixture-cal. (a=0.25)†	.192	.213	.530 (Qwen3-VL-32B)
Mixture-cal. (a=0.5)	.197	.211	.530 (Qwen3-VL-32B)
Merit-robust [31]	.222	.231	.358 (Llama)
Robust (Thm. 3.5)	.231	.231	**.231** (6/9 tight)
Robust, share $\le$ 0.25	.197	.211	.530 (Qwen3-VL-32B)
Oracle (per-curve)	.192	—	.200
Other indices, human-calibrated/robust			
MAD, ESCI (↓)	.060/.132	.108/.132	.251/.132
AWRF, TREC'22 (↑)	1.00/.999	1.00/.999	.992/.997
Gini, Steam (↓)	.000/.005	.014/.007	.053/.014
P-MMF share, Steam (↑)‡	.38	.42	.26 (Qwen3-VL-32B)

5.3 Cost of Serving All Readers

Cost decomposition. Table 2 reports the cost decomposition. Consider an oracle that optimizes for one realized curve whose identity is known. Its worst-type unfairness exceeds the human-only floor by only 0.000–0.053 (column $U_{\text{orc}}^* - U_H^*$), which is the part of the premium attributable to agent presence alone. The remaining 74–100% of every domain's premium comes from serving heterogeneous curves at once. Absolute premiums span 0.031 to 0.209. Across

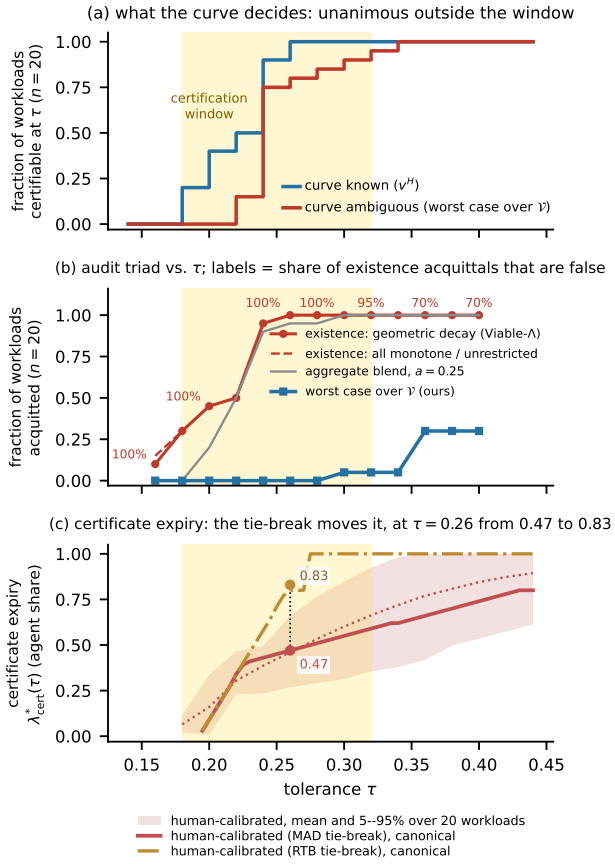


Figure 3: Twenty bootstrap ESCI workloads with curves stepped in 1/20. (a) Fraction of workloads certifiable at tolerance τ with a known or ambiguous curve. The verdicts differ only inside the shaded window. (b) Existence audits over geometric-decay, monotone, and unrestricted families certify a strict superset of the workloads certified by the worst-case audit. All of these certificates are false where the worst-case audit certifies no workloads, and 70–95% are false where it does; the monotone and unrestricted families differ from the geometric one on at most one workload. (c) Certificate expiry $\lambda^*_{\text{cert}}(\tau)$ for the human-calibrated policy: mean and 5–95% band over the 20 workloads, and the canonical workload under the MAD and robust tie-breaks. At $\tau = 0.26$ the tie-break alone moves expiry from agent share 0.47 to 0.83.

domains, this cost reflects the need to serve several realized instruments simultaneously rather than the distinction between human and agent traffic. Moving agent curves closer to the human model need not remove the premium while heterogeneity remains, nor does knowing the aggregate agent share, because the platform still faces multiple vertices. Table 3 reports the joint LP over the 200-query ESCI workload, and Figure 4 sweeps the agent share.

Policy inversion. The human-calibrated policy is optimal for humans but 176% worse for its worst agent type. On the three

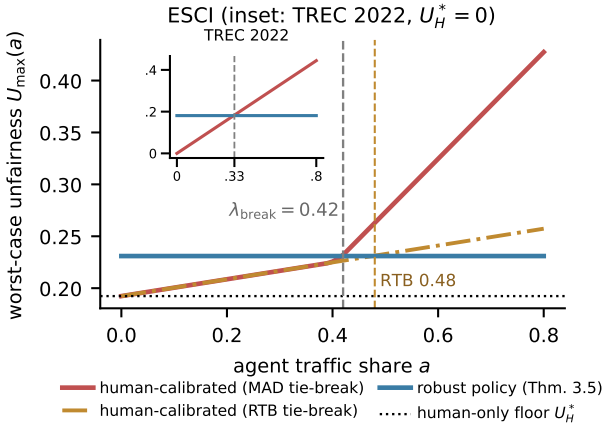


Figure 4: Worst-case unfairness versus agent share on ESCI. The inset shows TREC 2022, where $U^*_H = 0$ and the whole floor is the premium. Unfairness under the human-calibrated policy increases convexly with the MAD tie-break and nearly linearly with RTB, while it remains constant under the robust policy. The policies cross at λ_{break}.

domains where exact fairness is attainable under the human curve, it reaches 0.25–0.56 under its worst agent curve (Table 2). The comparison with the relevance-sort baseline illustrates this effect. It has higher unfairness than the human-calibrated policy under the human curve (0.287 against 0.192) but lower unfairness under the worst agent curve (0.464 against 0.530). Here a fairness intervention calibrated to one curve increases unfairness for the least-served agent reader relative to no intervention. Optimization under one curve can use allocation flexibility that has no fairness cost under that curve while concentrating exposure for readers with different positional weights. Fairness for one reader therefore does not imply fairness for all readers, demonstrating a practical consequence of the audit-logic failure.

Failed heuristics. Calibrating to the average curve performs poorly everywhere, even for humans. Calibrating to the known blend at $a = 0.25$ or 0.5 leaves worst-type unfairness unchanged relative to the human-calibrated policy because optimizing a single mixture does not control unfairness at the vertices (Jensen). The blend-optimal set contains a human-optimal point and the common tie-break selects it, so those rows of Table 3 are distinct policies that coincide on every reported value. A well-calibrated point estimate of the instrument does not provide guarantees over the full set.

Robustness cost. The robust policy holds every type at or below U^*, with six of the nine curves (the human curve and eight agent curves) tight. Its premium over the human-only optimum is 0.039, and its worst-type unfairness is within 0.031 of the oracle's. The binding provider is never the smallest. On the utility axis, fairness costs 3–11% of NDCG@10 against relevance sorting, and robustness to curve ambiguity adds at most 0.005 NDCG in every domain (Table 2). The additional utility cost of robustness is therefore small, despite the observed unfairness under the human-calibrated policy. Under the MAD, AWRF, and Gini indices, the human-calibrated

Table 4: What moves the reported quantities: perturbations of the measured set on the canonical ESCI workload. Premium, break-even, and sign-indeterminate providers of 15 under the human-calibrated policy; nulls over 20 draws and seeds over 3, mean±sd. The access/choice comparison is a separate re-solve whose choice baseline is .036/.41/4; the human-side set adds up to +.009 on the other domains.

Curve set	premium	λ_{break}	s-ind.
Measured curves (baseline)	.039	.42	4
Direct K=10 re-measurement, 3 seeds	.050±.003	.47±.01	4±0
Real ESCI items (7 of 8 re-measured)	.037	.61	4
Access curves in place of choice	.051	.38	3
TV ball ε=0.05 around every curve	.064	.39	5
Position-permuted curves (null)	.047±.009	.56±.09	4.5±0.5
Dirichlet random curves (null)	.017±.009	.13±.10	7.7±1.8
Leave one curve or one family out	.033–.039	.40–.42	4
Human side as a set (DCG, $\text{RBP}_{.8}$, $\text{RBP}_{.9}$)	.039	.42	6
Mixture cap $\bar{w} = 1/4$ on any agent curve	.038	.50	4

policy likewise deteriorates most under the worst curve, while the robust policy is nearly curve-invariant (Table 3, lower block).

Break-even and share bounds. For the human-calibrated policy, $U_{\max}(a)$ crosses the robust policy's flat line at $\lambda_{\text{break}} = 0.42$ on ESCI, and across domains the break-even ranges from 0.13 to 0.42 (Table 2). Below this threshold, the robust policy incurs a premium for protection against higher agent shares; above it, the robust policy dominates. The deployed optimum shifts the threshold because fairness LPs are degenerate, and a solver-vertex optimum of the same LP reaches break-even at 0.32. A platform confident that $a \le \bar{a}$ can instead solve the same LP over $\text{conv}(\{v^H\} \cup \mathcal{A}(\bar{a}))$, whose binding vertices are v^H and the endpoint mixtures $(1-\bar{a})v^H + \bar{a}\, v^{A_j}$. At $\bar{a} = 0.25$ it attains 0.211 at that share for 0.005 more under the human curve, but its worst type at full agent traffic returns to 0.530 (Table 3). The guarantee applies only up to the assumed bound.

What moves these numbers. The reported quantities depend on two inputs with different effects. The degree of allocative conflict determines the floor: binary merits or full-length lists permit exact fairness and eliminate the premium, while steeper merits enlarge it. The measured instrument instead fixes the curve-specific quantities, the break-even and which providers have undecidable signs. Table 4 collects every perturbation of the set: re-measuring, swapping access for choice, inflating, permuting, or dropping curves changes set membership and the break-even, from 0.13 under random curves to 0.61 under real items, while the premium stays within 0.03 of its baseline throughout. The set's membership changes, but distinct curves remain.

Existing remedies. A merit-robust LP [31], which protects perturbed merit targets under the known human curve, pays a premium comparable to ours at a=0 yet degrades under agent traffic (0.358 at a=1 versus our 0.231). Protecting against merit uncertainty incurs a premium without protecting against curve uncertainty. FELIX-style avoidance [14] cannot alter exposure here because exposure depends on the rank marginals alone. P-MMF [34], the max-min re-ranker used in our Steam preprocessing, guarantees each provider an exposure-count quota without modeling position,

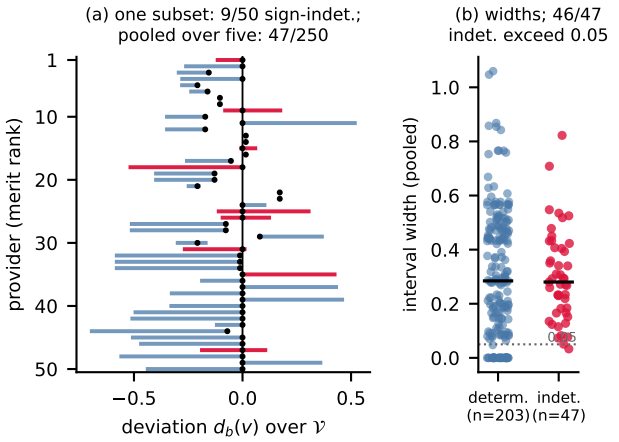


Figure 5: Provider-level incidence for the ESCI top-50 under the human-calibrated policy. (a) Merit-ranked deviation intervals over $\mathcal{V}$ for one of five independent 200-query subsets. Dots show values under v^H, red intervals cross zero, and 47/250 intervals cross zero across all subsets. (b) Interval widths pooled across subsets and split by sign determinacy. Of the 47 indeterminate intervals, 46 are wider than 0.05 and their median width is 0.28. The indeterminacy is therefore not explained by narrow interval widths.

so once exposure depends on position that guarantee inherits the curve (Table 3, P-MMF row). Bot detection separates human and agent traffic but does not identify which agent curve applies, even with perfect detection. Treating the human side as a set (DCG, $\text{RBP}_{0.8}$, $\text{RBP}_{0.9}$ [22]) or preventing an agent monoculture with a mixture cap has little effect on the floor (Table 4). The premium therefore reflects heterogeneity among reader curves.

5.4 Who Absorbs the Residual

Residual ambiguity remains under every policy and affects individual providers. Its visibility depends on the audit's aggregation level.

Sign identification. Under a known curve, every provider's deviation is a point. Under the ambiguity set, curve disagreement produces the entire interval width. The interval crosses zero for 27–73% of the top-15 providers in every domain under the human-calibrated policy (Table 2). Across five independent ESCI top-50 subsets, the rate is 19% under the canonical policy and 35% under the same LP's solver-vertex optimum (Figure 5). Sign-indeterminacy therefore depends on the deployed optimum, not only on the workload. The crossing intervals also have substantial width. Of the 47 crossing intervals, 46 are wider than 0.05 and have the same median width as the determinate intervals. Widths grow with merit depth, but crossings do not concentrate in the tail. Sign unidentification goes beyond uncertainty about magnitude: an affected provider cannot determine whether it is under-exposed.

Robust incidence. The robust policy neither randomizes within queries nor repositions providers. Instead, it changes cross-query coordination, and the two policies disagree on most per-query rank

mass. Each provider's exposure thereby becomes nearly curve-invariant. Mean interval widths shrink by more than half on ESCI and Steam, centers move toward zero, and sign-indeterminacy falls on three of four domains (Table 2, s-ind. H / R). Steam is the exception: the smallest premium coincides with the least improvement in sign identification. Robustness uses amortization slack [6], which becomes critical under ambiguity. The interpretation of the remaining zero-crossing intervals also differs. A human-calibrated crossing indicates potentially large harm of unknown direction, whereas a robust-policy crossing leaves only the direction unknown because the magnitude is certified at U^*.

Tie-break trade-off. Many policies attain U_H^* exactly, so deployment requires a tie-break whose choice affects provider-level audit outcomes. The robust tie-break (RTB) selects the exact stage-one optimum that minimizes $\max_{v\in\mathcal{V}} U(Pv)$ using one further LP. Without increasing human-curve unfairness, it reduces the worst case at full agent traffic from 0.530 to 0.274, compared with a robust floor of 0.231. It moves the break-even from 0.42 to 0.48 on ESCI and to 0.48–0.59 across all four domains (Table 2, RTB column), and extends every certificate's expiry (Figure 3c). However, recentering provider intervals near zero raises top-50 indeterminacy from 8/50 to 25/50. The MAD tie-break keeps indeterminacy lowest, while an arbitrary solver vertex provides neither benefit ($\lambda_{\text{break}} = 0.32$). The available degree of freedom can extend certificate lifetime or improve verdict centering, but not both.

Aggregation level. LastFM illustrates the effect of aggregation. At the demographic level of artist gender, the premium nearly vanishes at 0.007. At the artist level, the premium for the same workload is 0.209. This thirty-fold gap results from changing only the aggregation unit, with the curves held fixed. Coarse groups average curve disagreement away because readers who over-expose one artist under-expose another in the same group. At the individual-provider level, the full interval width remains. Most fairness benchmarks and disclosure regimes use group-level audits. Such an audit can correctly certify a platform by its own definition even when provider-level fairness is not identifiable, especially where reader disagreement is largest. The aggregation level therefore determines whether the audit captures or averages out provider-level differences. An audit that reports only group-level results should say so.

6 Discussion and Conclusion

Audit-logic failure. With AI agent readers, the instrument that converts rank into exposure becomes a set. The multiplicity is ontic rather than epistemic. Existence-style acquittal then becomes unsound because it asks whether some curve makes a ranking look fair while the platform serves all realized readers. The cost is determined by the need to serve readers simultaneously, and unresolved ambiguity affects individual providers. The empirical results trace the consequences of this logical failure.

Optimization versus auditing. Optimization cost and audit indeterminacy can differ. The premium reflects the optimizer's problem and has a low associated utility cost, whereas sign-indeterminacy describes the limits of the audit and is not always reduced by the robust policy. How soon a platform should abandon a human-calibrated policy depends on the deployed optimum, the value scale κ, and the protective norm: on one ESCI workload the break-even is 0.32, 0.42, or 0.48 for a solver vertex, the canonical MAD tie-break, or RTB. LP degeneracy creates a degree of freedom that can support robust operation or clearer ex-post attribution, but not both. A deployment should state which objective its tie-break serves. A disclosure regime that requires provider-level verdicts has reason to prefer the tie-break that keeps them determinate.

Audit procedure. First measure a curve set $\mathcal{V}$ under the deployed interface and item format, and disclose the serialization. Compute the worst-case envelope $U_{\max}(a)$ instead of searching for a curve that supports certification. Report the agent-share expiry $\lambda^*_{\text{cert}}(\tau)$ and the re-measurement tolerance from Proposition 3.4. Report provider verdicts as intervals whenever the sign is not identified, and state the aggregation level used. Beyond λ_{break}, solve the robust LP on the measured set and choose the tie-break according to the intended objective. A deployment may replace our stress-test curves with its own measured readers without changing this procedure.

Main implication. Serving multiple reader types simultaneously incurs a fairness premium, and residual ambiguity affects individual providers. A single replacement for the human curve does not account for heterogeneous agent readers. An audit designed for one reader model must change its logic when several models coexist. The measurements do not support relying on curve convergence: none of the eight curves we measure is close to the human model, and every re-measurement in Table 4 changed the set's members without eliminating heterogeneity.

7 Limitations

Measurement scope. Our measured curves are controlled stress-test instruments, not universal deployment estimates. They depend on serialization, model version, and item format. Section 4 shows that set membership changes while set-valuedness persists, so a deployment should re-measure $\mathcal{V}$ under its own interface and apply the same audit. In a preliminary comparison with six older or smaller models of the same families, not reported here, newer generations drifted toward flat without approaching the human model, so a measured set can change over time and should include its measurement date. Three main models are 4-bit, although quantization does not drive the shapes in the one measured contrast. GPT-5.6 curves were measured with reasoning off and are therefore lower bounds on decision-layer structure. GPT-5.6-sol was not re-measured on real items. Mistral-Small's label compliance also drops there, leaving its real-item curve with fewer scored trials.

Normative scope. We report κ=1, treating one agent choice as worth one human examination. Since v^H is an examination and v^A a choice probability, valuing an agent choice at κ human examinations converts the break-even to traffic share $\lambda/(\kappa(1-\lambda)+\lambda)$, so at $\kappa = 3$ the range 0.13–0.42 becomes 0.05–0.19. Our choice is conservative if an agent choice is closer to a conversion than a human examination ($\kappa > 1$), and optimistic only when an agent choice still awaits human approval ($\kappa < 1$). The scale moves the deployment threshold but not the verdict logic.

Modeling scope. The min–max formulation protects against any mixture in $\text{conv}(\mathcal{V})$, which is appropriate when agent composition is unobserved or known only as an interval. Better mixture

estimates can tighten the set through a share bound or a divergence ball around an estimated mixture. We treat the agent-share interval as an input rather than claiming to estimate it. Providers are non-strategic here, although a strategic provider could manipulate cross-query coordination or steer an interval across zero. Agent-oriented manipulation is already documented as generative-engine optimization [3]. The human side is a single examination model from 2002 (DCG); treating it as a set of standard decay models moves the robust floor by at most 0.009 (Table 4), but a deployment should substitute its own click-estimated curve. Merit is application-defined, and LastFM uses popularity, which is itself a function of past exposure. These choices affect the allocative geometry and therefore the premium, but not the failure of existence auditing under simultaneously realized reader instruments.

Ethical Considerations

All experiments use public benchmark datasets (ESCI, TREC Fair Ranking 2022, Steam, and LastFM-360K). No human subjects were involved, and no user data was collected. The LastFM artist-gender attribute describes public figures and is used only for aggregate exposure accounting, following its established use in fairness research [13]. Agent curves were measured on synthetic items through public APIs and open weights within their terms of use. The work is meant to strengthen auditing by showing how a common audit logic false-acquits under curve ambiguity and by supplying sound replacements. Two misuse risks arise. A platform could invoke ambiguity to resist accountability, but the worst case remains exactly computable, so ambiguity does not prevent a worst-case audit. Sign-indeterminacy could also be cited to dismiss provider complaints. We instead frame interval reporting as a disclosure obligation, not grounds for dismissal. Channel-specific policies can constitute cloaking, and a platform must assess their transparency before deployment. The robust policy also trades some consumer-side welfare for provider-side fairness. The paper quantifies these utility costs but does not choose the norm, aggregation level, or threshold.

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